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CMSC 204 Graphs Project 6 Writeup

1. What did you learn?

I learned how to utilize the concepts of graphs which was first introduced to me in a visual way in an IDE project setting using GUI in a real-world scenario (GPS and connecting towns with roads).

2. What issues did you encounter, if any?

I did not encounter any major issues with this project but it did take me a very long time to make sure all of the required methods were implemented properly.

3. What test cases did you run?

I thoroughly tested all function methods (outlined in the project description), every constructor, and a wide variety of edge/special cases I came up with. I also made sure to avoid retesting methods that were already covered by the public tests to avoid redundancy.

4. What were the expected and actual outputs (in the JavaFX GUI) when you ran it?

I expect that the GUI will function according to the GUI screenshots in the project description. I will provide the screenshots down below.

5. Will your project be able to pass a set of private test cases?

While I cannot say this for sure, I highly believe my project can pass a set of private test cases because I tested all my projects exactly the same in the way I mentioned for question #3 and so far all of my projects in CMSC 204 have passed all private test cases. I also did not

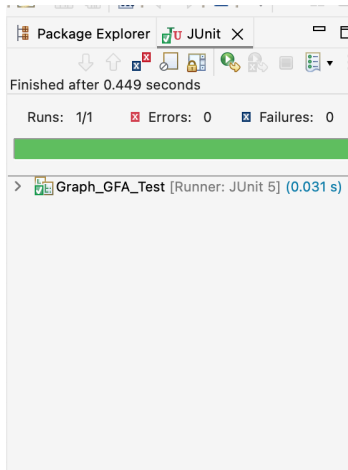
hardcode any part of my project in any way to artificially produce expected output.

6. Assumptions

I assumed this project would be fairly short given the relatively small number of classes I was required to implement and having experience with a similar project in CMSC 203 that was based on encrypting/decrypting strings.

JUnit Test Screenshots:

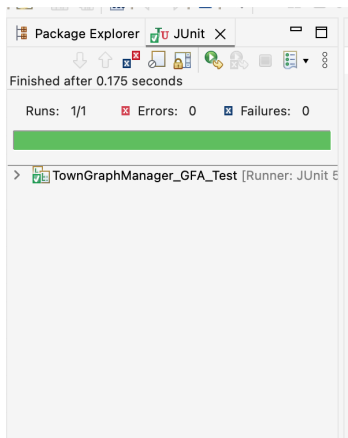
Graph_GFA_Test.java



The screenshot shows the JUnit test results for `Graph_GFA_Test`. The test was run by JUnit 5 and completed successfully in 0.031 seconds. The Package Explorer on the left shows the test class under the `CMSC 204 Project 6 Graphs` package. The console output indicates that the test finished after 0.449 seconds with 1/1 runs, 0 errors, and 0 failures.

```
1  
2  
3  
4 import static org.junit.Assert.*;  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17 public class Graph_GFA_Test {  
18     private GraphInterface<Town, Road> graph;  
19     private Town[] town;  
20  
21     @Before  
22     public void setUp() throws Exception {  
23         graph = new Graph();  
24         town = new Town[3];  
25  
26         for (int i = 0; i < 3; i++) {  
27             town[i] = new Town("Town_" + i);  
28             graph.addVertex(town[i]);  
29         }  
30  
31         graph.addEdge(town[0], town[1], 2, "Road_1");  
32     }  
33 }
```

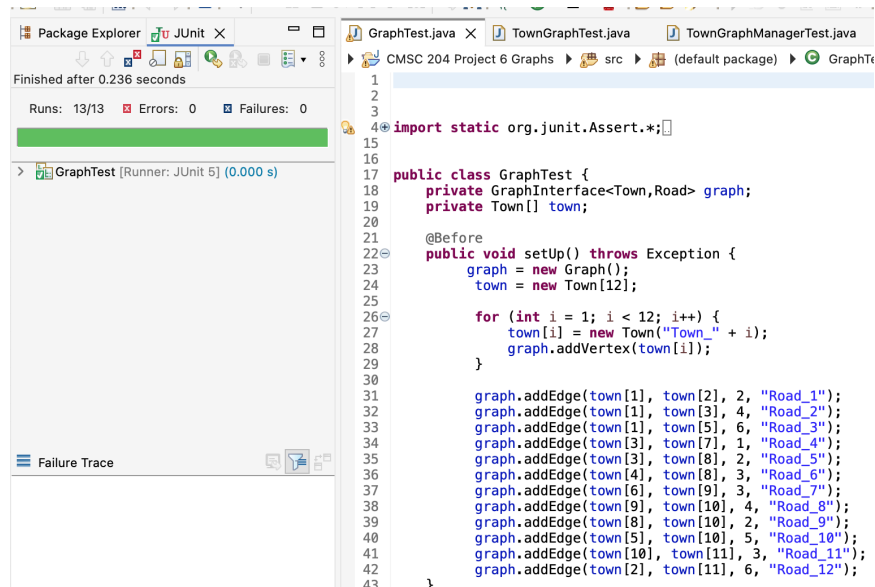
TownGraphManager_GFA_Test.java



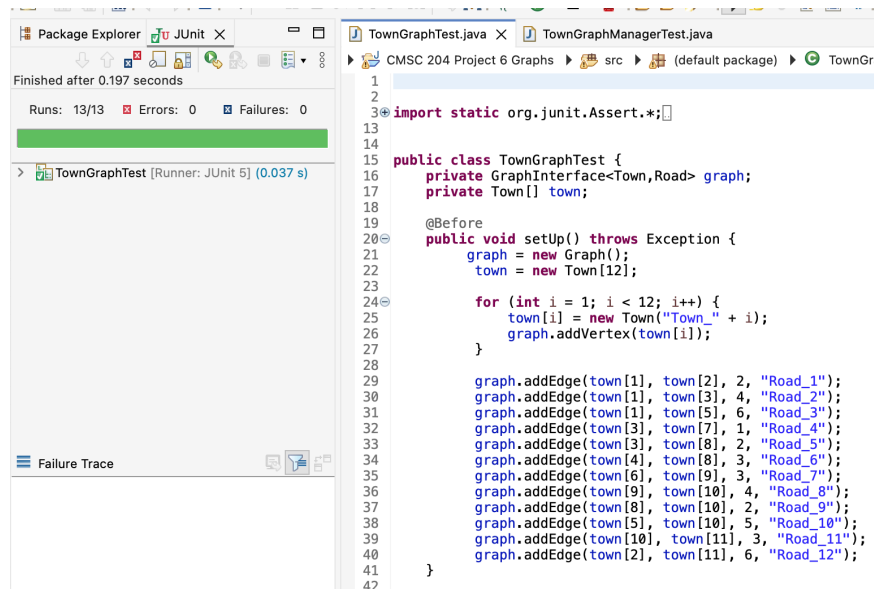
The screenshot shows the JUnit test results for `TownGraphManager_GFA_Test`. The test was run by JUnit 5 and completed successfully in 0.175 seconds. The Package Explorer on the left shows the test class under the `CMSC 204 Project 6 Graphs` package. The console output indicates that the test finished after 0.175 seconds with 1/1 runs, 0 errors, and 0 failures.

```
1  
2  
3  
4 import static org.junit.Assert.*;  
5  
6  
7  
8  
9  
10  
11  
12  
13 public class TownGraphManager_GFA_Test {  
14     private TownGraphManagerInterface graph;  
15     private String[] town;  
16  
17     @Before  
18     public void setUp() throws Exception {  
19         graph = new TownGraphManager();  
20         town = new String[12];  
21  
22         for (int i = 1; i < 12; i++) {  
23             town[i] = "Town_" + i;  
24             graph.addTown(town[i]);  
25         }  
26  
27         graph.addRoad(town[1], town[2], 2, "Road_1");  
28     }  
29 }
```

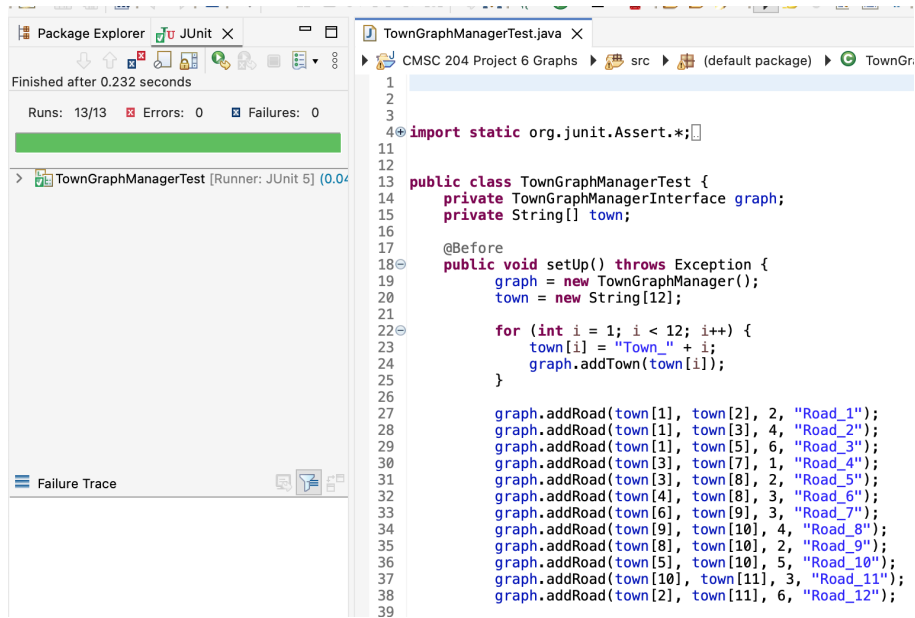
GraphTest.java (I assume this is the public tests for Graph.java)



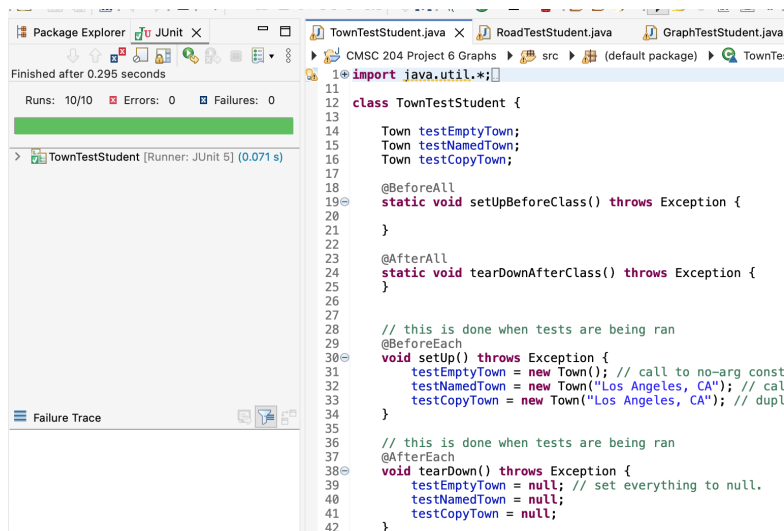
TownGraphTest.java (I assume this is a public test class)



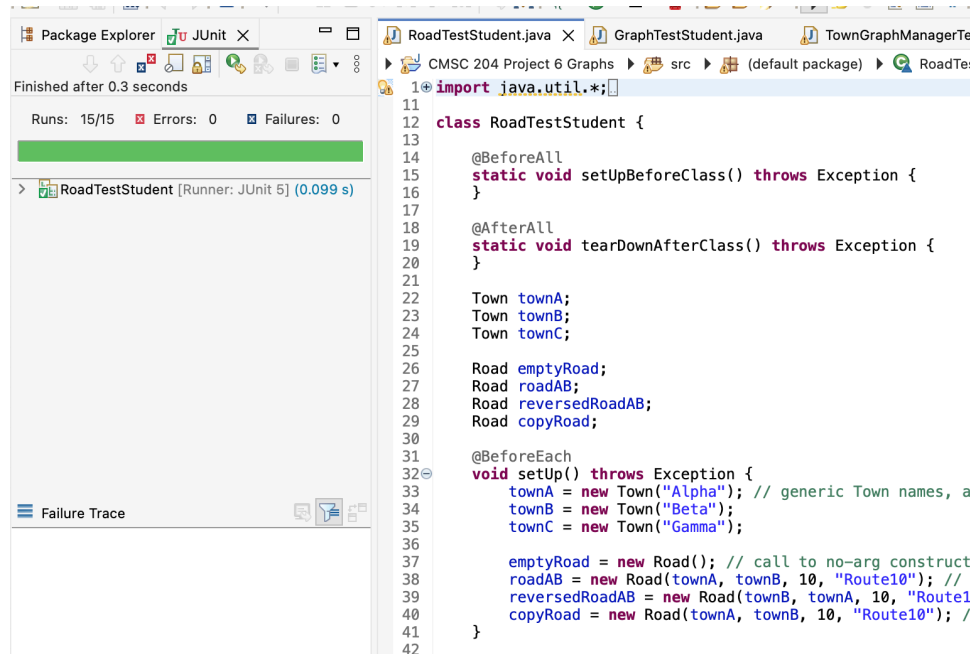
TownGraphManagerTest.java (I assume this is the public tests for TownGraphManager.java)



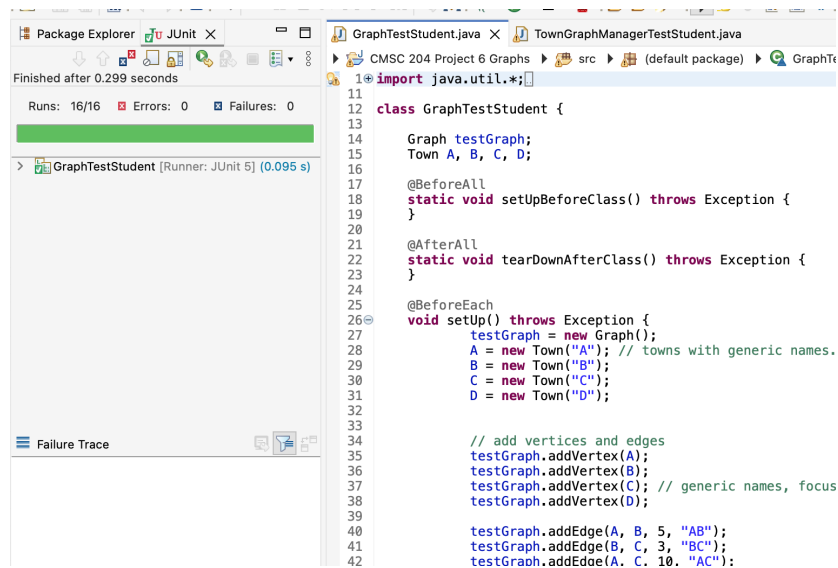
TownTestStudent.java



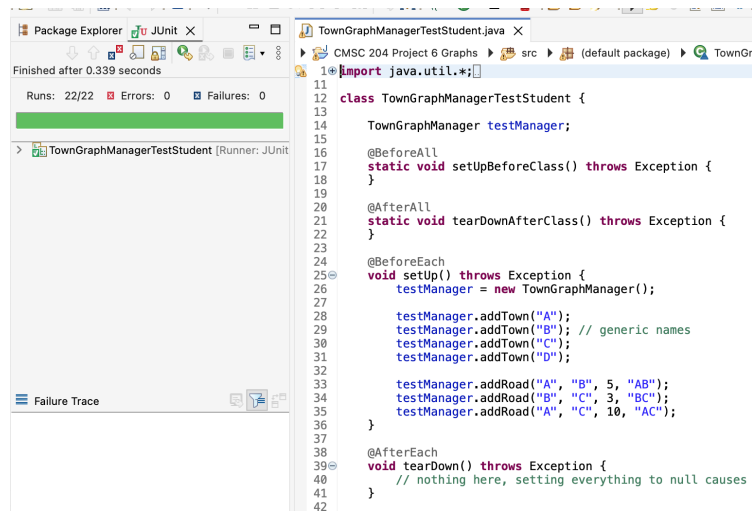
RoadTestStudent.java



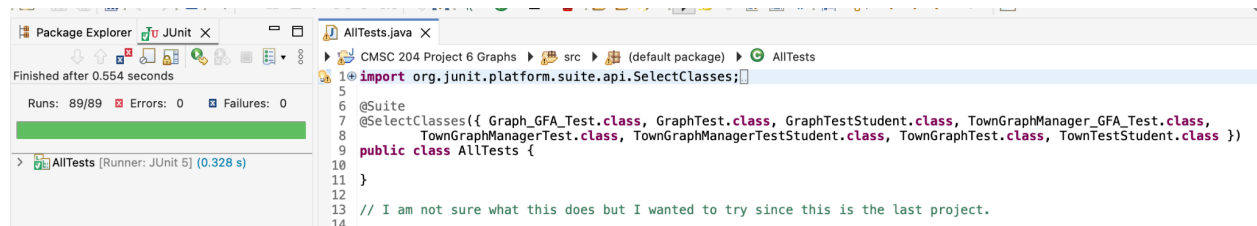
GraphTestStudent.java



TownGraphManagerTestStudent.java

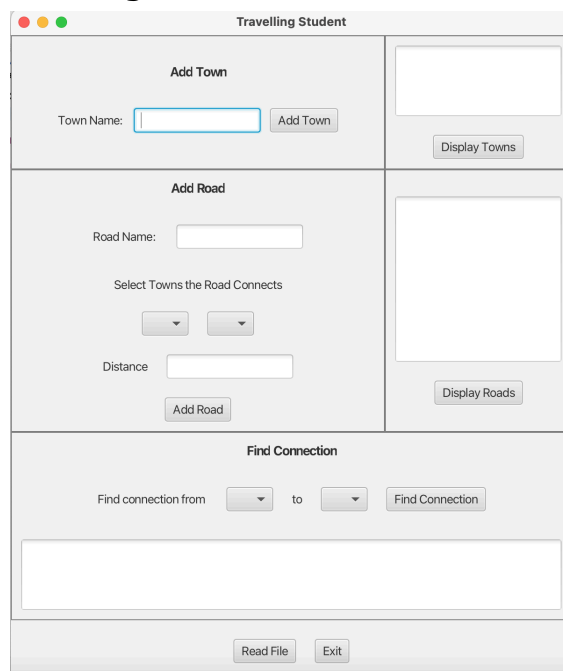


AllTests.java (JUnit Test Suite)



JavaFX Screenshots

Running the GUI



Adding Towns

Travelling Student

Add Town

Town Name:

Add Road

Road Name:

Select Towns the Road Connects

Distance

Find Connection

Find connection from to

Travelling Student

Add Town

Town Name:

Add Road

Road Name:

Select Towns the Road Connects

Distance

Find Connection

Find connection from to

Finding a Connection

The screenshot shows the 'Travelling Student' application window. It has three main sections: 'Add Town', 'Add Road', and 'Find Connection'. In the 'Add Town' section, 'Brooklyn, NY' and 'Queens, NY' are listed in a text area. In the 'Add Road' section, 'testRoad' is listed. In the 'Find Connection' section, the 'Find connection from' dropdown is set to 'Brooklyn, NY' and the 'to' dropdown is set to 'Queens, NY'. The 'Find Connection' button is highlighted. Below the dropdowns, a text area displays the result: 'Brooklyn, NY via testRoad to Queens, NY 15 mi'. At the bottom of the window are 'Read File' and 'Exit' buttons.

Travelling Student

Add Town

Town Name:

Brooklyn, NY
Queens, NY

Add Road

Road Name:

Select Towns the Road Connects

Distance

testRoad

Find Connection

Find connection from to

Brooklyn, NY via testRoad to Queens, NY 15 mi

No Connection

The screenshot shows the 'Travelling Student' application window. In the 'Add Town' section, 'Brooklyn, NY', 'Osaka, Japan', and 'Queens, NY' are listed. In the 'Add Road' section, 'testRoad' is listed. In the 'Find Connection' section, the 'Find connection from' dropdown is set to 'Brooklyn, NY' and the 'to' dropdown is set to 'Osaka, Japan'. The 'Find Connection' button is highlighted. Below the dropdowns, a text area displays the result: 'You can't get there from here'. At the bottom of the window are 'Read File' and 'Exit' buttons.

Travelling Student

Add Town

Town Name:

Brooklyn, NY
Osaka, Japan
Queens, NY

Add Road

Road Name:

Select Towns the Road Connects

Distance

testRoad

Find Connection

Find connection from to

You can't get there from here

Blank Road

Travelling Student

Add Town

Town Name:

Brooklyn, NY
Osaka, Japan
Queens, NY

Error

Road name cannot be blank

Road

Distance

Brooklyn

Find Connection

Find connection from to

Brooklyn, NY via testRoad to Queens, NY 15 mi